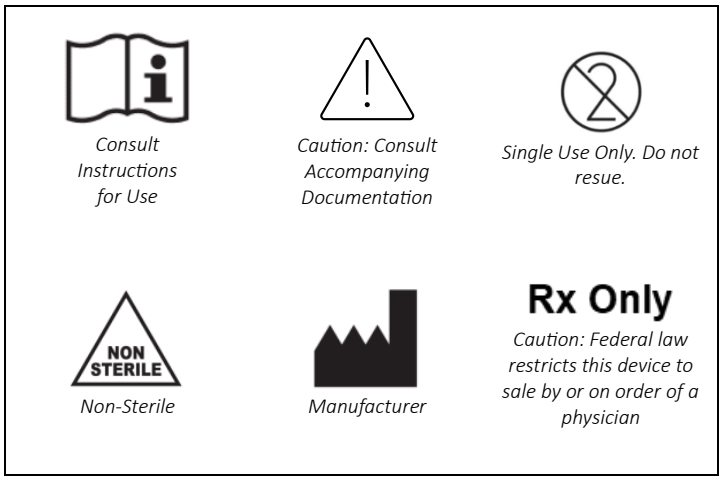
ELEVATION SPINE

Saber-XA System

Instructions for Use

SYMBOL TRANSLATION



Consult Instructions for Use

Caution: Consult Accompanying Documentation

Single Use Only. Do not resue.

Non-Sterile

Manufacturer

Rx Only

Caution: Federal law restricts this device to sale by or on order of a physician

ELEVATION SPINE

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BEFORE USING PRODUCT, READ THE FOLLOWING IMPORTANT INFORMATION

This booklet is designed to assist in using the Elevation Spine, Saber-XA System. It is not a reference for surgical techniques.

CAUTION: Federal law (USA) restricts this device to sale by, or on the order of, a physician.

GENERAL DESCRIPTION: The SABER-XA System is an integrated fixation lumbar interbody and anterior lumbar plate system comprised of a collection expandable lumbar interbodies, anterior plates and fixation implants. The SABER-XA expandable interbody includes additively manufactured endplates and traditionally manufactured internal components; the anterior plate and fixation implants are traditionally manufactured. The interbody, plate and fixation implants are available in an assortment of dimensional combinations to accommodate the individual anatomic and clinical circumstances of each patient.

The basic shape of the SABER-XA interbody is a structural column to provide surgical stabilization of the spine. Each interbody is comprised of additively manufactured external structure, having a roughened surface of 9µm, whereas the internal structure is comprised of traditionally machined and anodized mechanism components. The inferior/superior aspects of the interbody incorporate multiple vertical cavities which can be packed with bone graft material. Each interbody is preassembled and provides the ability to independently control height and lordotic expansion and may accept up to two bone screws or bone anchors.

INDICIATIONS FOR USE:
The Elevation Spine SABER-XA System is indicated for intervertebral fusion in patients with DDD at one or two contiguous levels from L1 to S1. These DDD patients may also have up to Grade 1 Spondylolisthesis or retrolisthesis at the involved levels. DDD is defined as discogenic back pain with degeneration of the disc confirmed by history and radiographic studies. These patients should be skeletally mature and have had six months of non-operative treatment. The SABER-XA Spacer is to be filled with autograft bone and/or allogenic bone graft composed of cancellous and/or corticocancellous bone.

The Elevation Spine SABER-XA Anterior Lumbar Plate, when used with screws and without the SABER-XA spacer component, is intended for anterior screw fixation to the lumbosacral spine (L1-S1) to treat instability resulting from: fracture (including dislocation and subluxation), tumor, degenerative disc disease (as defined above), pseudoarthrosis, spondylolysis, spondylolisthesis, scoliosis, kyphosis, lordosis, spinal stenosis, or failed previous spine surgery. The Anterior Lumbar Plate is not to be used alone with Spikes, butmay be used in construct with the spacer and spikes.

All interbody fusion constructs of SABER-XA System are intended to be used with FDA cleared supplemental fixation.

CONTRAINDICATIONS: The SABER-XA System contraindications include, but are not limited to:

1. Active systemic infection, infection localized to the site of the proposed implantation, or when the patient has demonstrated allergy or foreign body sensitivity to any of the implant materials.
2. Signs of local inflammation.
3. Prior fusion at the level(s) to be treated.
4. Severe osteoporosis, which may prevent adequate fixation.
5. Conditions that may place excessive stresses on bone and implants, such as severe obesity or degenerative diseases, are relative contraindications. The decision whether to use these devices in such conditions must be made by the physician considering the risks versus the benefits to the patient.
6. Patients whose activity, mental capacity, mental illness, alcoholism, drug abuse, occupation, or lifestyle may interfere with their ability to follow postoperative restrictions and who may place undue stresses on the implant during bony healing and may be at a higher risk of implant failure.
7. Any patient not willing to cooperate with postoperative instructions.
8. Any condition not described in the indications for use.
9. Fever or leukocytosis.
10. Patient has known metal sensitivity.
11. Pregnancy.
12. Any other condition that would preclude the potential benefit of spinal implant surgery, such as the presence of tumors or congenital abnormalities, fracture local to the operating site, elevation of sedimentation rate unexplained by other diseases, elevations of the white blood count (WBC), or a marked left shift in the WBC differential count.
13. Any case not needing a fusion.
14. Patients with a known hereditary or acquired bone friability or calcification problem should not be considered for this type of surgery.
15. These devices must not be used for pediatric cases or where the patient still has general skeletal growth.
16. Any case where the implant components selected for used would be too large or too small to achieve a successful result.
17. Any case that requires the mixing of metals from two different components or systems.

18. Any patient having inadequate tissue coverage at the operative site or inadequate bone stock or quality.
19. Any patient in which implant utilization would interfere with anatomical structures or expected physiological performance.

WARNINGS AND PRECAUTIONS: One of the potential risks identified with the system is death. Other potential risks which may require additional surgery include:

1. Mixing of dissimilar metals can accelerate the corrosion process. Stainless steel and titanium implants must NOT be used together in building a construct.
2. The SABER-XA System devices should be implanted only by surgeons who are fully experienced in the use of such implants and the required specialized spinal surgery techniques. Prior to use, surgeons should be trained in the surgical procedures recommended for use of these devices.
3. The SABER-XA anterior plate is not intended for screw attachment or fixation to the posterior elements (pedicles) of the cervical, thoracic, or lumbar spine.
4. The correct selection of the implant is extremely important. The potential for success is increased by the selection of the proper size, shape and design of the implant. Based on the dynamic testing results, the physician should consider the levels of implantation, patient weight, patient activity level, other patient conditions, etc., which may impact on the performance of the device.
5. These devices are provided as single use only implants and are not to be reused or reimplanted regardless of an apparent undamaged condition.
6. The SABER-XA System is used to augment the development of a spinal fusion by providing temporary stabilization. If fusion is delayed or does not occur, material fatigue may cause breakage of the implant. Damage to the implant during surgery (i.e., scratches, notches) and loads from weight bearing and activity will affect the implant's longevity.
7. The correct handling of the implant is extremely important. Use care in handling and storage of devices. Store the devices in a clean, dry area away from radiation and extreme temperatures and corrosive environments such as moisture, air, etc.
8. Patients with previous spinal surgery at the level(s) to be treated may have different clinical outcomes compared to those without a previous surgery.
9. Components of this system should not be used with components of any other manufacturer.
10. Potential risks identified with the use of this system, which may require additional surgery, include: device component breakage, loss of fixation/loosening, non-union, vertebral fracture, neurologic, vascular or visceral injury.

These warnings do not include all adverse effects that could occur with surgery in general but are important considerations to orthopedic implants. General surgical risks should be explained to the patient prior to surgery. Use this device as supplied and in accordance with the handling and use information provided.

POTENTIAL ADVERSE EVENTS: Prior to surgery, patients should be made aware of the following possible adverse effects in addition to the potential need for additional surgery to correct these effects:

1. Loosening, bending or breakage of components
2. Displacement/migration of device components
3. Tissue sensitivity to implant material
4. Potential for skin breakdown and/or wound complications
5. Non-union or delayed union or mal-union
6. Infection

7. Nerve damage, including loss of neurological function (sensory and/or motor), paralysis, dysesthesia, hyperesthesia, paresthesia, radiculopathy, reflex deficit, cauda equina syndrome
8. Dural tears, cerebral spinal fluid leakage
9. Fracture of vertebrae
10. Foreign body reaction (allergic) to components or debris
11. Vascular or visceral injury
12. Change in spinal curvature, loss of correction, height and/or reduction
13. Urinary retention or loss of bladder control or other types of disorders of the urogenital system
14. Ileus, gastritis, bowel obstruction or other types of gastrointestinal system compromise
15. Reproductive system compromise including impotence, sterility, loss of consortium and sexual dysfunction
16. Pain or discomfort
17. Bursitis
18. Decrease in bone density due to stress shielding
19. Loss of bone or fracture of bone above or below the level of surgery
20. Bone graft donor site pain, fracture, and/or delayed wound healing
21. Restriction of activities
22. Lack of effective treatment of symptoms for which surgery was intended
23. Need for additional surgical intervention
24. Death

MRI SAFETY INFORMATION: The SABER-XA System has not been evaluated for safety and compatibility in the MR environment. It has not been tested for heating, migration, or image artifact in the MR environment. The safety of the SABER-XA System in the MR environment is unknown. Scanning a patient who has this device may result in patient injury.

PACKAGING: The implants and instruments are provided nonsterile and are steam sterilized prior to use, as described in the STERILIZATION section below. Following use or exposure to soil, instruments and instrument trays and cases must be cleaned, as described in the CLEANING section below.

HANDLING AND USE

All instruments and implants should be treated with care. Improper use or handling may lead to damage and/or possible malfunction. Products should be checked to ensure that they are in working order prior to surgery. All products should be inspected prior to use to ensure that there is no unacceptable deterioration such as corrosion (i.e. rust, pitting), discoloration, excessive scratches, notches, debris, residue, flaking, wear, cracks, cracked seals, etc. Non-working or damaged instruments should not be used, and should be returned to Elevation Spine.

Implants are single use devices and should not be cleaned. Re-cleaning of single use implants might lead to mechanical failure and/or material degradation. Discard any implants that may have been accidently contaminated.

CLEANING

Instruments should be thoroughly cleaned separately from instrument trays and cases. Lids should be removed from cases for the cleaning process, if applicable. All instruments that can be disassembled must be disassembled for cleaning. All handles must be detached. Instruments may be reassembled following sterilization. The products should be cleaned using neutral cleaners before sterilization and introduction into a sterile surgical field or (if applicable) return of the product to Elevation Spine.

Cleaning and disinfecting can be performed with aldehyde-free solvents at higher temperatures. Cleaning and decontamination must include the use of neutral cleaners followed by a deionized water rinse. Note: certain cleaning solutions such as those containing formalin, glutaraldehyde, bleach and/or other alkaline cleaners may damage some devices, particularly instruments; these solutions should not be used.

The following cleaning methods should be observed when cleaning instruments and instrument trays and cases after use or exposure to soil, and prior to sterilization:

1. Immediately following use, ensure that the instruments are wiped down to remove all visible soil and kept from drying by submerging or covering with a wet towel.
2. Disassemble all instruments that can be disassembled.
3. Rinse the instruments under running tap water to remove all visible soil. Flush the lumens a minimum of 3 times, until the lumens flush clean.
4. Prepare Enzol® ® (or a similar enzymatic detergent) per manufacturer's recommendations.
5. Immerse the instruments in the detergent and allow them to soak for a minimum of 2 minutes.
6. Use a soft bristled brush to thoroughly clean the instruments. Use a pipe cleaner for any lumens. Pay close attention to hard to reach areas.
7. Using a sterile syringe, draw up the enzymatic detergent solution. Flush any lumens and hard to reach areas until no soil is seen exiting the area.
8. Remove the instruments from the detergent and rinse them in running warm tap water.
9. Prepare Enzol® ® (or a similar enzymatic detergent) per manufacturer's recommendations in an ultrasonic cleaner.
10. Completely immerse the instruments in the ultrasonic cleaner and ensure detergent is in lumens by flushing the lumens. Sonicate for a minimum of 3 minutes.
11. Remove the instruments from the detergent and rinse them in running critical water for a minimum of 2 minutes.
12. Dry instruments using a clean soft cloth and filtered pressurized air.
13. Visually inspect each instrument for visible soil. If visible soil is present, then repeat cleaning process starting with Step 3.

Instrument Case/Tray Automated Cleaning Procedure
The following steps should be completed in sequence:

1. Prepare a neutral pH enzymatic cleaning solution using lukewarm water as per the manufacturer's recommendation.

2. Fully immerse the devices and allow to soak for a minimum of two (2) minutes.

3. Following the soak time, flush any lumens of the device using a syringe.

4. Rinse the devices under lukewarm running water for a minimum of one (1) minute, while agitating the devices. Agitation includes actuating all movable parts, such as opening and closing hinges and moving the devices around under the running water.

5. Use a clean soft bristled brush and/or pipe cleaner to brush and aid in the rinse for the exterior and interior of device components. Use a syringe to flush any lumens.

6. Place the devices back into the designated locations of the case/tray and load the case/tray set into an automated washer (Getinge 86-Series, Steris 444, or equivalent).

7. The washer cycle parameters are as follows:

Phase	Recirculation Time (min.)	Water Temperature	Detergent
Pre-Wash 1	2:00	Cold Water	N/A
Wash 1	4:00	Hot Water	Enzymatic Cleaner
Wash 2	3:00	60°C (140°F)	Mild Detergent
Rinse 1	2:00	Hot Water	N/A
Dry	7:00	N/A	N/A

a) Utilize OptiPro® or Enzol® enzymatic cleaner or equivalent.

b) Utilize Prolystica® 2X mild detergent or equivalent.

c) Only purified water that has been filtered using deionization, reverse-osmosis, or ultra-filtration should be used for all final rinsing.

d) Visually examine to determine if all adherent visible soil has been removed. Repeat the cleaning procedure if visible debris is detected.

STERILIZATION: The SABER-XA System implants and instruments are provided non-sterile and must be sterilized prior to use. Recommended sterilization methods include steam autoclaving after the removal of all protective packaging and labeling. Prior to sterilization, verify that all instruments are in their open and unlocked position within the instrument tray(s). The use of an FDA cleared sterilization wrap is recommended. The following validated steam autoclave cycle has been validated to an SAL of 10^{-6}.

Steam - Pre-Vacuum Cycle
Temperature 132°C (270°F)
Cycle Time: 4 Minutes
Minimum Dry Time: 60 Minutes
Double Wrapped

INSTRUCTIONS FOR USE: The Elevation Spine SABER-XA System should only be implanted by surgeons who are fully experienced in the use of such implants and the required spinal surgery techniques. Refer to the SABER-XA Surgical Technique Guide (651-40002) for complete instructions for use. For product information or to obtain a copy of the surgical technique guide, please contact an Elevation Spine representative by phone (844) 415-0226.

 INSTRUCTIONS:

PREOPERATIVE:

1. Preoperative instructions to the patient are essential. The adverse effects, warnings, precautions and limitations should be understood by the surgeon and explained to the patient prior to the surgery.
2. Only patients that meet the criteria described in the indications should be selected.
3. Correct selection of the implant is extremely important. An adequate inventory of sizes should be available at the time of surgery.
4. Patient conditions and/or predispositions such as those mentioned in the Contraindications, Precautions and Warnings should be avoided.
5. The surgeon should be familiar with the use and handling of all components and instruments of the system prior to surgery.

6. Proper function of the surgical instruments and components should be verified prior to every surgical procedure. All instruments and components must be sterilized before use.

INTRAOPERATIVE:

1. The primary goal of this surgery is to arthodese selected vertebrae. Adequate exposure, bony preparation and crafting are essential for this result.
2. The placement of the SABER-XA System devices should be checked radiographically.
3. Care should be taken when positioning the implants to avoid neurological damage. Extreme caution should be used around the spinal cord and nerve roots.

POSTOPERATIVE:

1. Adequately instruct the patient on postoperative care, use and limitations and potential complications. Successful healing depends on postoperative care and the patient's ability and willingness to follow instructions.
2. The patient must be made aware of the limitations of the implant and that physical activity and load bearing may cause premature loosening, bending or fracture of the internal fixation device. The patient should be warned to avoid falls, sudden jolts, mechanical vibrations, and lifting, twisting motions and restrict any type of sport participation. An active, debilitated, or uncooperative patient who cannot properly restrict activities may be at particular risk during postoperative rehabilitation.
3. If a nonunion develops, or if the implants loosen, fracture, corrode, migrate, cause pain, or stress, the device(s) should be evaluated, revised and/or removed. Patients with evidence of these conditions should be closely observed, the possibilities of further deterioration evaluated, and the benefits of reduced activity, revision or removal considered.
4. Periodic X-rays for at least the first year postoperatively are recommended to detect any evidence of nonunion, changes in position, loosening, bending or cracking of components.
5. Any retrieved devices must never be reused under any circumstances.

PRODUCT COMPLAINTS: The customer or health care provider should report any dissatisfaction with the product quality, labeling or performance to Elevation Spine immediately. Elevation Spine should be notified immediately of any product malfunction by telephone, fax or written correspondence. When filing a complaint, the name, part number and lot number of the part should be provided along with the name and address of the person filing the complaint.